Micropolis Online

Constructionist Educational Open Source SimCity

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Micropolis Project Goals

- Develop a fun web based educational game, based on the original SimCity source code.
  - Focused on constructionist education.
  - Built out of free open source software.
  - Runs on the server, plays in the browser.
  - Supports collaborative, democratic, multi-user play.

Accomplishments So Far

- Ported and shipped the TCL/Tk/X11 version of Micropolis on the One Laptop Per Child XO-1.
- Developed a cross platform, desktop version of Micropolis, based on GTK/Python.
- Developed an open source, internationalized, web based version of Micropolis.
  - Client based on OpenLaszlo/Flash on the client.
  - Server based on TurboGears/Python/C++.
- Renovated, re-factored and documented the code.
- Established a solid groundwork and open framework for developing an educational, collaborative, multi-player game

Open Source Software

- Micropolis is built with a carefully chosen, well proven, widely supported set of open source, cross platform tools and libraries.

- Micropolis itself and its TileEngine module are open source (GPL3). The client runs on the proprietary Adobe Flash player. The client will also run in standard browsers with a little more work, thanks to OpenLaszo.

- Linux. Apache. MySQL. Python. Cairo. Pango. GTK. GCC. SWIG. PyCairo. PyGTK. TurboGears. Genshi. SQLAlchemy. CherryPy. PyAMF. AMFast. Pie menus. OpenLaszlo. Red5. Java. Tomcat. WikiMedia. PHP. Subversion. Doxygen.

Web Server and Client

- Server: Linux, Apache, MySQL, Python, TurboGears, C++
- Portable, cross platform C++ and Python code.
  - Linux and Unix
  - Mac OS/X
  - Windows (not currently maintained but viable)
- Deploys on standard Fedora Linux server, i.e. Amazon Elastic Computing Server, etc.
- Client: OpenLaszlo, XML, JavaScript, AMF
- Web based client runs in Adobe Flash player, widely

Internationalization

- All text comes from Unicode XML localization files, so the text can be translated to support different languages and countries.
- Currently supports English and Dutch, but other languages are easy to add, given translation resources.
- Web based tool for conveniently reviewing and editing translations.
- This is a very important feature, since we're initially delivering to the Dutch educational system, and we intend to support many other languages in the future.

Multi Player Collaboration

- Designed, developed and shipped a multi-player X11 version of SimCity in 1993 for DUX software, demonstrated in Amsterdam at INTERCHI '93.

- Micropolis has been systematically redesigned with multi-player collaboration and constructionist education in mind.

- Currently laying the groundwork for multi player and online community support, while getting the single player game up and running first.

- User accounts. Saved cities. Shared cities. Chat. Robots and Avatars. Basic MediaWiki integration. Client/server session, messaging and simulation engine plumbing.

Constructionist Education

- The original SimCity was an exemplary constructionist educational game, which has been successfully used in many classrooms, but it was a closed commercial product, not specifically designed to support education.

- Since Micropolis is open source SimCity, we can open up, renovate, visualize, document and extend it in wonderful ways, to specifically support constructionist education, social networking and collaborative democratic play.

- "One of the most wonderful possibilities about this venture is that it will bring together very fluent designers from many worlds of computing (more worlds than usually combine to make a game) in the service of the children. We should really try to pull this off!" -Alan Kay

Educational Applications

- Encourage creative writing by blogging, twittering, chatting, debating, writing and publishing newspaper articles about events and issues in the city.

- Teach democratic processes. Writing proposals. Campaigning for support. Discussing and debating issues. Voting on bills.

- Publish wiki pages with interactive annotated maps, timelines of events and creative writing about the cities.

- Open up the simulator to enable plug-in programmable zones, tools, robots, avatars, events, policies, courseware, etc.

- Teach older kids to program Python and C++. Implement a visual behavior programming language (like Robot Odyssey and SimAntics). Teach younger kids to think logically and program robots (like Logo turtles and Lego Robotics).

Next Steps

- Shared city library. Ratings. Tagging. Journals. Discussions. Linking. History browsing. Sharing. Voting. Profiles. Friends.

- Text, voice and video chat. In-game chat with avatars on the map. Integrated video player for help, tutorials and lessons. Integrated html frame for synchronized web browsing.

- MediaWiki integration, to support creative writing by students and courseware development by teachers.

- Facebook integration. Having your friends vote on issues to fund your grant proposals. Write a proposal to convince five of your friends to fund your proposal to build an airport.

- PayPal integration. Donations, subscriptions and micro-payments, for virtual money, cheats, high speed simulation, extra storage, fund raising, etc.

Google Wave

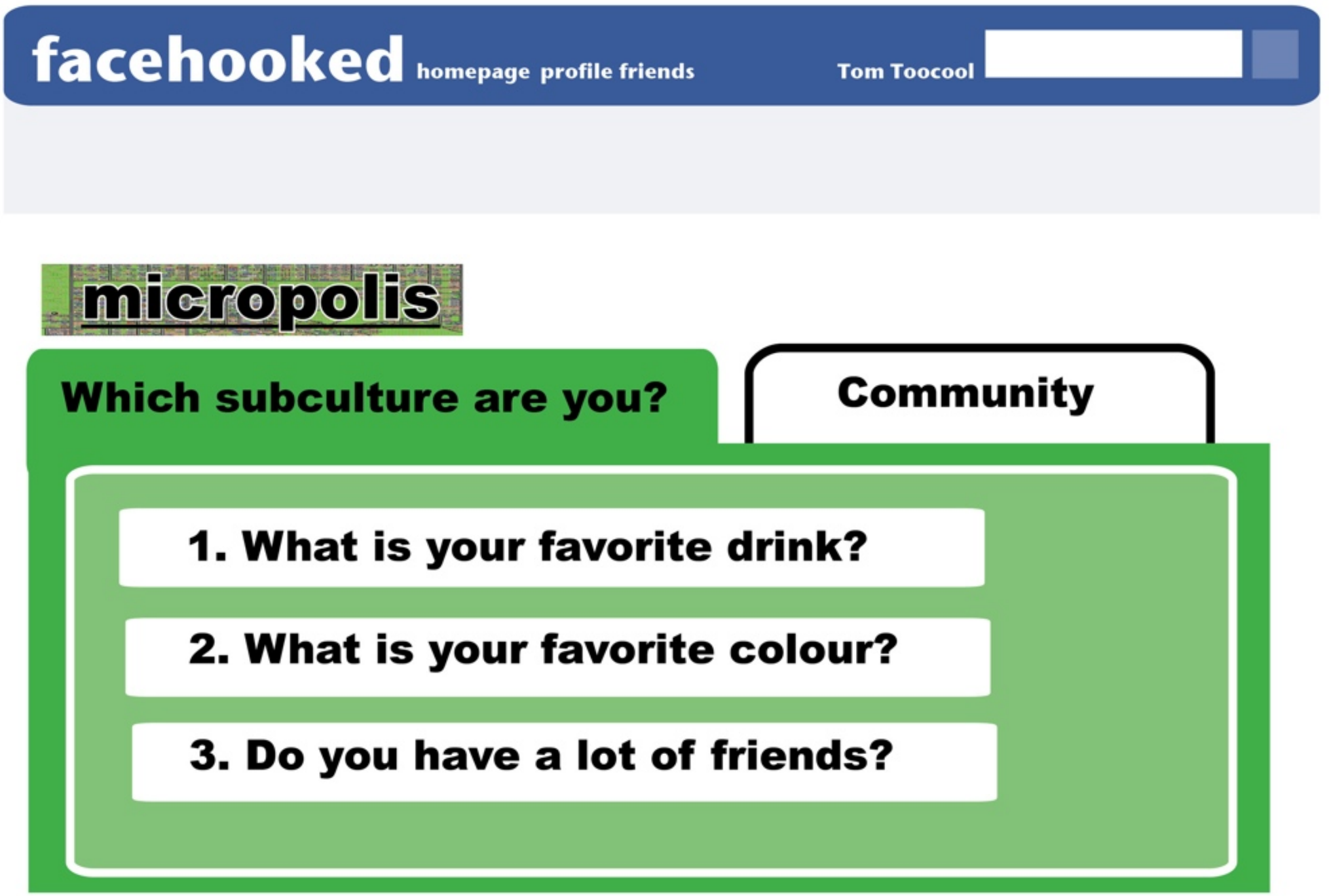
- Wave is an ideal platform for Micropolis, because it supports discussion, storytelling, collaborative editing, timeline playback, group decision making, and democratic processes (multi player voting).

- Wave Gadgets: Micropolis client runs in Flash player embedded in a Wave. Multiple views in different users' browsers of the same shared simulation running on the Micropolis server.

- Wave Robots: Micropolis server joins a wave, and creates and updates content in the wave (publishing stories about events in the game that players can edit and interact with).

- Constructionist Education: Teach language skills, interpersonal communication, politics, proposal writing, as well as the obvious social studies, science, urban design, etc.

Identity


facehooked
homepage profile friends
Tom Toocool
micropolis
Which subculture are you?
Community
1. What is your favorite drink?
2. What is your favorite colour?
3. Do you have a lot of friends?

Voting

facehooked homepage profile friends Tom Toocool

micropolis

Invite friends| go to city| visit other city| micropolis.com

Tom Toocool

Amsterdamned, 1243.55 population, $12.0040 , 34 visitors

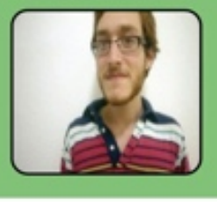
Voting | Citizens | Journal | Community



Citizens vote: 65% YES







Citizen


facehooked
homepage
profile
friends
Tom Toocool
micropolis
Invite friends| go to city| visit other city| micropolis.com
Tom Toocool
Amsterdamned, 1243.55 population, $12.0040 , 34 visitors
Voting
Citizens
Journal
Community
Suggestions
Citizens
Visitors

Journal


facehooked
homepage profile friends
Tom Toocool
Tom Toocool
Wall
Info
Pictures
Video
FIRE! FIRE! THERE IS NO FIRE BRIGADE?
CITIZENS ARE FURIOUS!
We need more education.
Lets have a vote.
JOURNAL;
FRIENDS DEMAND VOTING: FIRE PROTECTION
NEW CHURCH
TAXES ARE UP
more...
So what are doing wanna go out Tnight?
big kiss and hughs me ;)
Friends

Community

facehooked homepage profile friends Tom Toocool

Flying Spaghetti Monster

Wall | Info | Pictures | Video



DEMONLISHED CITIZENS ARE AFFRAID!



COME SEE MY CITY! AMSTERDAMNED



MTV COMMUNITY, MUCH BETTER!

MUSIC TELEVISION

Community






COMMUNITY IS GROWING:
POPULATION: 34004.0038
RANK: 4

Chat


Note: 1-8-2004, 13:34
Note: 1-8-2004, 13:34
Note: 1-8-2004, 13:34
zones.
Hi Tom, wassup?
What' s going on here?
Allright allright, i have
the best city ever...

Squatters


Note: 1-8-2004, 13:34
Note: 1-8-2004, 13:34
zones.
Warning! Squatters found !
Anonymous visitors. They
want to take over your city.
See www.wikipedia.com/squat